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Exploring Roadside Management Optimisation Opportunities: A Systematic Review on Arc Routing Problems Applications and Modelling

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Abstract

The gradual growth of road infrastructure worldwide has heightened the importance of sustainable roadside management for preserving adjacent ecosystems and maintaining critical ecosystem services. While the Arc Routing Problem (ARP) presents a promising framework for optimising operations carried out along an arc, its application to roadside management remains unexplored despite its advantages over traditional routing problems.

This study presents a comprehensive systematic literature review to analyse how ARPs have been applied across various real-world scenarios and to identify relevant features for roadside management applications. The presented analysis is conducted in two stages. First, a bibliometric analysis introduces the context of the ARP scientific domain. Then, a deeper analysis is based on a detailed examination of model characteristics, network configurations, and sustainability challenges across different application domains.

The findings reveal a predominance of theoretical work with few real-life applications but highlight significant opportunities for adapting existing ARP frameworks to roadside management, particularly in integrating sustainability criteria and local context considerations. However, crucial gaps in current ARP applications are identified, notably the limited incorporation of sustainable and regenerative practices in model formulation. This review provides a structured decision support framework for researchers and practitioners, outlining specific directions for developing ARP models tailored to sustainable roadside management while emphasising the need to balance operational efficiency with sustainable objectives.

Keywords: routing problem, arc routing problem, roadside management, optimisation, systematic literature review

1. Introduction

The relationship between road infrastructure and economic development is widely studied, with an estimated 25 million kilometres of new roads projected to be built by 2050 [70]. However, this rapid expansion has a non-negligible impact on the surrounding territories, which can be controlled and mitigated by the vegetated strips of land adjacent to roads, known as roadside or road verges.

Roadside management entails maintaining the vegetated areas next to roadways and enhancing security for road users, crucial for biodiversity and ecosystem services [96]. Practices include mowing, pruning, routine maintenance, and ecological enhancement to maintain habitat quality [84]. Therefore, operational challenges such as planning maintenance to minimise traffic disruption and efficiently allocate resources, are crucial [83]. Climate change further complicates these tasks, requiring adaptive strategies that accommodate increased storm intensity and shifting vegetation patterns [79].

Managing maintenance tasks under a sustainable and regenerative approach can tackle these challenges by not only mitigating social and environmental considerations, but also actively restoring and enhancing ecosystem services [97, 44]. While sustainability focuses on minimising harm, a regenerative approach goes forward and aims to create net-positive outcomes, such as increasing biodiversity, improving soil quality and supporting socio-ecological development through roadside ecosystems [48].

Despite these benefits, decision-makers face significant challenges in defining roadside maintenance strategies, planning operations, optimising routes, and allocating resources to minimise operating costs while coping with sustainable and regenerative objectives [12]. Therefore, transitioning to a regenerative model of roadside management requires long-term projects tailored to local contexts, integrating comprehensive strategies that address the complexities of road infrastructure while actively contributing to ecological resilience and community wellbeing [99, 44].

These challenges in implementing sustainable and regenerative management have been emphasised in the roadside literature, particularly by [96] and [44], who identify the development of models for roadside management scenarios as a crucial area for future research. They underscore the necessity of sustainable planning research that offers a holistic, long-term assessment of ecosystem services provided by roadsides, including their potential to regenerate natural capital and socio-ecological synergies.

Addressing these future research needs, routing problems emerge as pivotal solutions for integrating operational planning with efficient logistical frameworks in roadside management. Among the most studied models is the Vehicle Routing Problem (VRP), which focuses on finding the most efficient routes for a fleet of vehicles to serve a set of customers located at specific nodes within a network [88]. On the other hand, the Arc Routing Problem (ARP) extends the principles of the VRP by focusing on servicing arcs and edges within a network [27]. By addressing the limitations of the VRP, the ARP provides decision-makers with a more suitable framework to optimise routing strategies in processes such as roadside management.

Emphasising this topic, the comprehensive literature review carried out by [27] provides a thorough overview of general ARP, covering their historical development, theoretical aspects, and practical applications, highlighting key variants and emerging topics. According to the authors, the main challenges on this topic lie in the need for novel algorithms, hybrid approaches, and advanced metaheuristics to address the resolution of existing models more efficiently, as well as the development of new ARP variations that can be adapted to the emerging applications.

Although the VRP has been the subject of more extensive study, the literature on ARPs has only recently begun to grow due to its usefulness in different applications in real-life problems. Nevertheless, further research is required to ascertain the applicability of ARPs in a range of emerging applications, including the growing focus on regenerative development [106]. In addition, to the best of our knowledge, a significant gap remains regarding the application of ARP models to roadside management. Despite its potential, there is currently no work that directly addresses roadside management challenges through this approach, let alone by integrating sustainable and regenerative aspects in modelling techniques.

Therefore, in order to guide future research in the design of ARP models for this critical application, it is now necessary to go deeper into how the proposed models have been developed for each specific application in the literature, which features have been considered for these purposes, and how they have been adapted to meet the unique requirements of each case.

To tackle this issue, this work carries out a systematic literature review of the applications of the ARP, assessing the relationships between the different real-life applications studied and relevant criteria identified in the literature, such as type of model, type of network and model features considered. Thus, this review aims to provide a decision support tool to guide future research to further develop the field of application of ARP for roadside management issues.

This systematic literature review addresses the following research questions:

1. How are studies of ARP applied to real problems developed in contrast to the ARP literature in general?
2. Which fields of application have been studied, and how have variations on classical ARP allowed modelling to be adapted to each problem?
3. How have sustainability aspects been included in the ARP literature applied to real-life problems?
4. Which solution methods are used to tackle complexity in ARPs applied to real-life problems?
5. What considerations should be explored to design an ARP model adapted for application in roadside management?

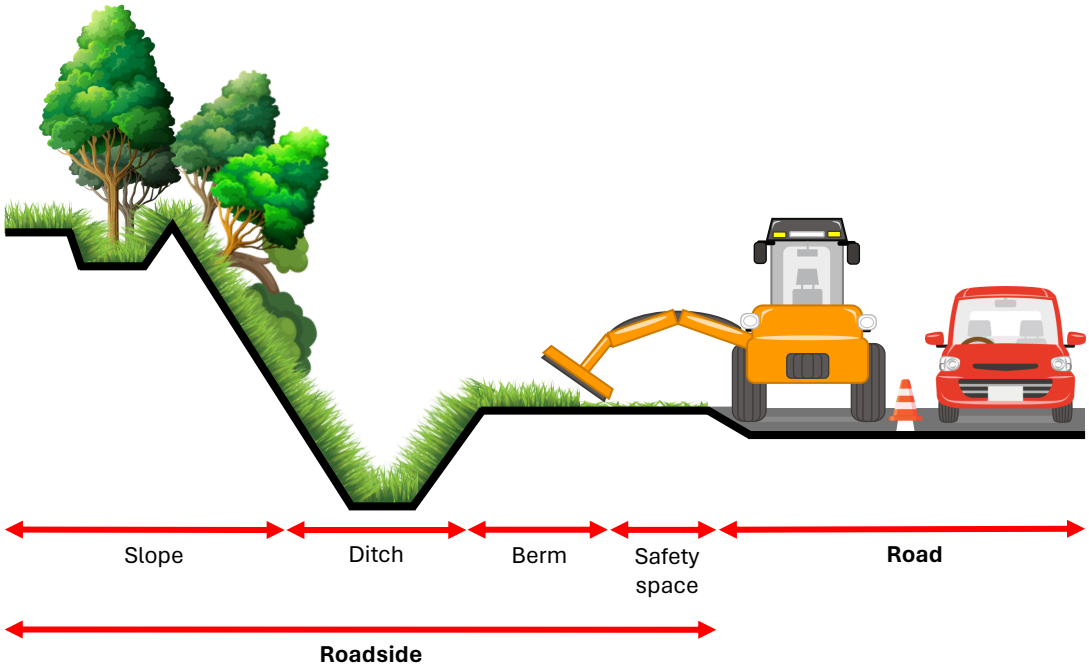


Figure 1: Road infrastructure illustration.

This review article is organised as follows. For a better understanding of the contents covered in this paper, a theoretical framework on roadside management issues and routing problems is provided in Section 2. In Section 3, the details of the process for reviewing articles according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) are presented. A descriptive analysis of the data found in the selected dataset is performed in Section 4. Then, a study and analysis of the evaluation criteria extracted from the literature is discussed in Section 5. In Section 6, the results of the previous analyses are summarised and assessed for determining the main needs for the application of roadside management in ARP models. Finally, conclusions and future directions for the designing of ARP models and their applications are further discussed as the results of this paper in Section 7.

2. Theoretical framework

In this section, a theoretical framework on the two main topics of this paper, roadside management and routing problems, is provided in order to offer a better understanding of the concepts that will be discussed in the following sections.

2.1. Roadside management

Roadside management involves the systematic maintenance of vegetation strips along the edges of roads [58]. The organisation of this activity differs from country to country, but in general, an agency is responsible for the maintenance of an administrative division (region, department, etc.), depending on their territorial conditions and priorities [79]. Each agency has a limited budget to acquire resources such as personnel, vehicles, and specific machinery, which must be assigned to work centres located within the territory to distribute the work.
The planning of roadside maintenance depends on several territory variables (vegetation cutting height, maintenance period, maintenance frequency, maintenance width, maintenance differentiation, biomass collection) [32, 93, 79]. Based on these decisions, roadside worksites are planned in order to determine the optimal routes for current vehicles, which better comply with these sustainable objectives. Special consideration is given to the logistics of moving heavy machinery and equipment between different segments of the roadside efficiently and safely [95, 68].

2.1.1. Road infrastructure

The composition of roadsides varies according to the country where they are located, the surrounding landscape and the specific road they border [96]. The main composition of roadsides can be described by three main elements: the berm, the slope, and the ditch [79]. The berm's main purpose is to guarantee road maintenance and user safety [136]. The slope forms a transition with the surrounding landscape [58]. In some cases, a ditch is also included between the berm and the slope, which collects and drains water from the road and adjacent areas, preventing flooding and ensuring smooth traffic flow [12]. Fig. 1 illustrates the composition of the road infrastructure described.

2.1.2. Main activities in roadside maintenance

Maintenance of roadside areas includes a variety of activities devoted to maintaining herbaceous and woody vegetation, each requiring specific machinery and techniques [95]. The specific tasks to be carried out depend on the type of management practised in each country. However, the most frequent activities in roadside management can be generalised as follows:

The initial step in most roadside management operations is worksite safety, which ensures security during maintenance activities through proper signage and barriers to inform and protect road users and workers [80].

Once security is granted, the operations in charge of maintaining the green areas are carried out. Mowing is one of the most common maintenance activities to be performed periodically. It allows the control of herbaceous vegetation height, which helps to maintain visibility and road safety [83, 112]. This activity utilises arm mowers or tractor-mounted mowers capable of handling different terrains and vegetation densities [79].

In addition, pruning and tree trimming are vital for keeping roadside trees and shrubs in check, preventing branches from obstructing road signs and lights, and reducing the risk of falling limbs [66]. This requires machines similar to those used in mowing activities, while tree pruning requires lifting platforms and hand tools.

Thirdly, biomass recovery and transportation are critical aspects of sustainable management, as they involve gathering plant material from mowing and pruning activities [32, 101]. Herbaceous and woody roadside biomass can be recovered and transported to recovery facilities [133, 33]. This activity requires additional equipment dedicated to collecting biomass into trailers and transporting it by truck [133, 78].

In countries where the road infrastructure includes a ditch (See Fig. 1), ditch cleaning is essential for ensuring proper drainage and preventing water accumulation, which could lead to road damage or safety hazards [79, 58]. Excavators and specially equipped trucks are used to remove silt, debris, and vegetation, particularly critical before and after rainy seasons to maintain effective drainage [79].

Overall, the roadside maintenance activities are very much related to the natural setup of each urban area that needs maintenance and is highly dependent on current equipment that the local authority needs to use but also on staff availability, waste collection goals, etc.

2.1.3. Sustainable and regenerative practices

The literature identifies different maintenance strategies, but under a sustainable management approach, the main objectives are to reduce maintenance costs, time and emissions, improve water and soil quality, provide safety for vehicles and travellers, improve the overall driving experience, and provide habitat for wildlife populations [58].

In addition, biomass management plays a crucial role in regenerative practices, but this decision can lead to both advantages and disadvantages. Waste extracted from roadside mowing (grass and wood) can be integrated into energy supply chains, as beneficial feedstock for use in biogas systems, anaerobic digestion, composting, or combustion, generating added value to these processes [82, 12]. However, this practice adds supplementary machinery investments and operational costs. Moreover, cutting too much can degrade soil quality and affect species in adjacent ecosystems [96]. Conversely, insufficient cutting on certain roads can reduce visibility, increasing the risk of accidents. Not collecting biomass, in turn, can enrich the soil, but in certain areas, this can promote the growth of invasive species. Therefore, it is essential to evaluate how much to cut and recover based on multiple scenarios that enable a sustainability assessment.

2.2. Routing problem models

A routing modelling problem suitable for roadside management needs to be done by integrating all these aspects, which can be challenging. A further deep dive into the routing modelling approach that can be employed is provided in the upcoming section.

There are two primary types of routing problems commonly discussed in the literature: the VRP and the ARP. VRP focuses on optimising the routes of a fleet of vehicles dispatched from one or several depots to serve a set of customers represented by nodes [27]. It has been extensively studied and applied in contexts where the delivery locations are discrete points demanding services or goods, such as in urban delivery and service systems [124].

In contrast, ARP deals with servicing requirements not at nodes, but along the arcs of a network [88]. This type of problem is particularly relevant in scenarios that involve routing a set of vehicles to cover specified arcs where each arc represents a task that needs to be performed, such as inspecting or maintaining the infrastructure. The challenge in ARP comes from the need to cover all required arcs as efficiently as possible, often while returning to the point of origin, making it crucial for applications that involve widespread geographical areas with specific service requirements along routes rather than at discrete locations [28]. Fig. 2 shows a basic representation of these two problems.

2.2.1. Model variations

There are several specific types of ARP that address different logistical challenges. The three main variations of the general ARP are described below.

- The Chinese Postman Problem (CPP). Often considered the simplest form of ARP. It seeks a minimum-length closed path or circuit that visits every arc at least once [37].

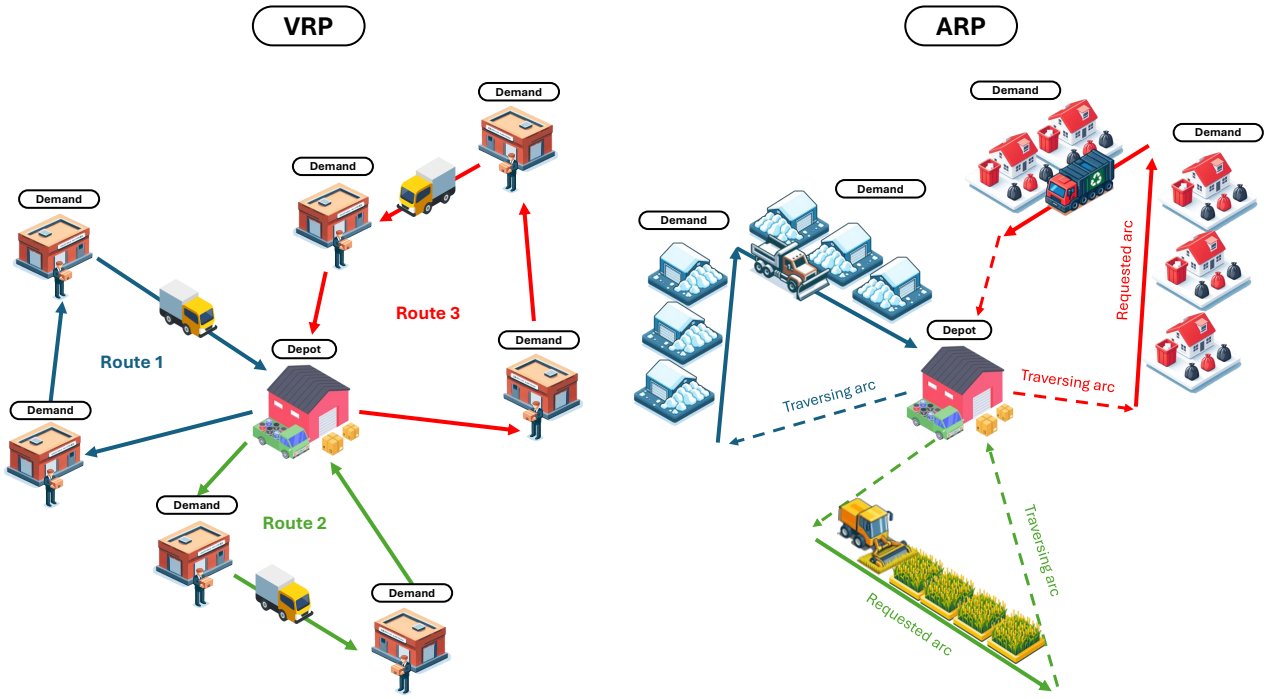


Figure 2: Representation of VRP and ARP networks.

- The Rural Postman Problem (RPP). Similar to the CPP but more specific. It focuses on finding a minimum-length route that covers only a specified subset of arcs in a graph [13].
- The Capacitated Arc Routing Problem (CARP). In this variation, each vehicle has a capacity constraint, which it cannot exceed when performing tasks like collecting waste or delivering goods [7].

This wide range of variations of the original problem has allowed the creation of models better adapted to the needs of different real-life applications.

2.3. Roadside Management modelling in Arc Routing Problems

To the best of our knowledge, despite the several applications studied in the literature, there is still no work that has studied roadside mowing management by applying the ARP.
The main difference between proper modelling of roadside management and ARP applications in the literature is the role of sustainable management, which mainly affects the objectives to be evaluated and the nature of the model's demand. According to the description provided in Section 2.1, a multi-objective approach is required in order to assess different sustainable objectives, finding the most suitable biomass management plan. To achieve this, the decision variables will depend not only on the binary decision of whether to cut or not, but also on how much to cut in each segment, based on the evaluation of each objective function. Thus, the model's demand can be defined as the amount of biomass required to cut, and it will vary according to the objective function. This feature has not been addressed in any other application in the ARP literature.
Additionally, according to the same description provided before, an ARP model for roadside management might include the following features:

- A directed graph in which the arcs representing roads include multiple sections according to the structure of the roadside, leading to a variable condition within the same arc.
- Multiple depots representing the work centres, and their respective location decisions.
- Strategic and operational decisions that optimise the limited budget for this activity, such as resource allocation, vehicle routing and workers' scheduling.
- A heterogeneous fleet of vehicles for each maintenance activity. These vehicles can vary in costs, operation time, and type of task.
- A multi-period planning.

Although each of these features has been studied in the literature, very few have modelled a problem including all of them. The lack of a model that includes all the characteristics described above (including the sustainable approach and the nature of the demand) motivates our current study and in-depth literature review.

3. Systematic literature review protocol

In this section, the protocol followed for the systematic literature review proposed is explained step by step. The PRISMA method is the review approach applied to organise and analyse the literature for this article [94]. This method was chosen because it is well-structured and organised for reporting systematic reviews with clear objectives, and its checklist and flow diagram have proved useful for several studies in the past. Table 1 details the steps to be followed for the selection of articles. The scientific domain searched corresponds to scientific publications related to the design of ARP models for real case applications. Then, the selection process for reviewing the literature is illustrated in Figure 3.

Stage	Principle	Description
Search Strategy	Type of studies	Journals and conference papers
	Keywords	Arc Routing Problem, real-life applications
	Search equation	(("*Arc Routing*" OR "*ARP*" OR "*Postman Problem") AND ("case study" OR "application" OR "real-life"))
	Period	Not limited
	Databases	Web of Science, Scopus and SpringerLink
Study selection	Criteria	1. Articles related to the analysis or evaluation of ARP. 2. Articles that model real case applications for the ARP.
	Procedure	1. Title, abstract, and keywords are screened. 2. Full article is reviewed. 3. Selection is made based on selection criteria.
Data extraction	Application and model characteristics	1. Real-life application 2. Network approach 3. Type of ARP model 4. Vehicle features 5. Time features 6. Nature of demand 7. Sustainability dimensions 8. Solution approach

Table 1: Systematic protocol for the literature review investigation.

3.1. Data extraction

The first step focuses on identifying relevant literature from online databases and several library resources, such as websites, organisation publications, and citation searching. The databases used in this stage were Web of Science, Scopus, and SpringerLink. Since this work aims to evaluate the changes in the literature through time, the publication date period was not limited. This way, it is possible to identify patterns of change in conjunction with the other characteristics evaluated. From this process, 411 documents were collected from several online databases, and 57 works were found from other resources (35 articles were found on websites, and 22 articles were found based on citation searching). In this step, the selected literature was primarily screened by removing duplicates, where 137 works were removed. At this point, these 274 publications screened are analysed in order to provide a context for the scientific domain searched. Then, a first filter was made by screening of publication's title, abstract, and keywords, where 142 articles were removed from databases and 33 from other resources because they were of a relevant low quality or out of the scope of this review article. After reducing the dataset obtained, a second filter was carried out by a full article reading, removing 55 publications. In this stage, since the focus is on the articles of application, the theoretical papers were removed, i.e., literature reviews, articles related to algorithm proposals, and models proposed without a particular field of application. Finally, the last step showed that the total number of literature works that were ensured to be included in this review article for an in-depth analysis was 101.

3.2. Evaluation criteria selection

For each selected article in the final dataset, data extraction was carried out by identifying evaluation criteria found in the literature that were relevant to the design of an ARP model adapted for roadside management (described in Section 2.3). The selected criteria are defined as follows.

1. **Field of application** for which the routing model is designed, including real-life application and type of operation (pick-up, delivery, other).

2. **Network approach** on which the problem is modelled. This means the type of graph, number of depots, and type of decisions (strategic, tactical or operational).

3. **Type of model** proposed - if it presents a variation of the original ARP model.

4. **Vehicles features** considered. This means the number of vehicles used (if this number is fixed or unlimited), if they have capacity constraints, and if the fleet is uniform (consisting of identical vehicles) or heterogeneous (differences in capacity, utility, or technology).

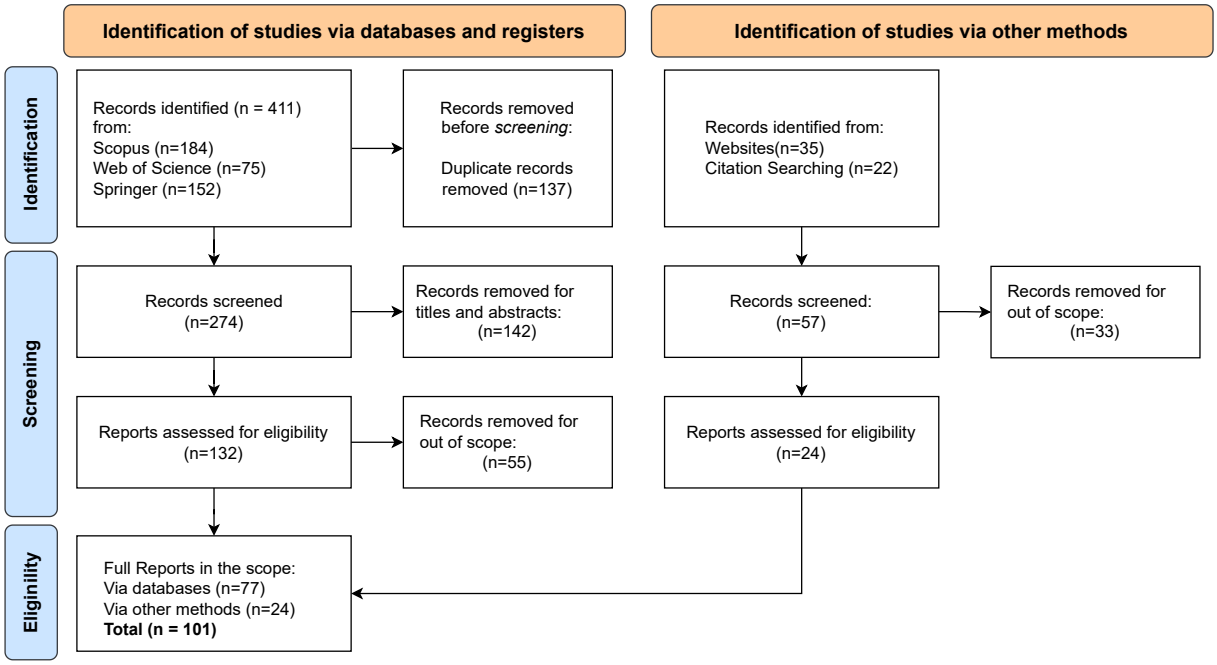


Figure 3: Flow diagram for the systematic review based on the PRISMA approach.

5. **Time features** according to the model needs, such as the planning period considered and time constraints in service time, working hours, traffic peak hours, among others.

6. The **Nature of the demand** - how it is calculated and which factors can affect it. For example, uncertainty, variability, other model parameters, etc.

7. **Sustainability dimensions** represented in the objective functions. To analyse these aspects, the approach provided by [11] has been followed. In this work, five dimensions have been proposed. In addition to the three traditional pillars: economic, environmental, and social, the political and technological dimensions are added for a complete multidimensional view.

8. **Solution approach** for the model programming. These methods play an important role in the development of ARPs since one of the main challenges of the current literature is the high-resolution times of these complex problems. Among these methods are exact methods, heuristics, and metaheuristics.

3.3. Results analysis process

Given the structure of the systematic review protocol, the documents selected will be processed and analysed in two stages. Fig. 4 presents a scheme of the methodology.

1. A **bibliometric analysis** is performed with the 274 records screened. This analysis has been used in research to handle large volumes of scientific data and to produce high research impact, in order to identify the emerging trends in a particular field [36]. In this stage, the distribution of articles through time, authors and keywords, is shown. In this way, the aim is to provide a broader perspective on the state of the art of ARP in general.

2. A **Modelling and applications analysis** of the selected criteria described above, is carried out. In this stage, the 101 selected papers are analysed and debated based on the chosen evaluation criteria. The objective is to identify how these criteria have been studied for the design of application models in this field and how this can be further applied to our current roadside management problem.

4. Bibliometric Analysis

To contextualise the state of the literature on ARPs, a descriptive analysis of the 274 articles screened in this research is carried out in this section.

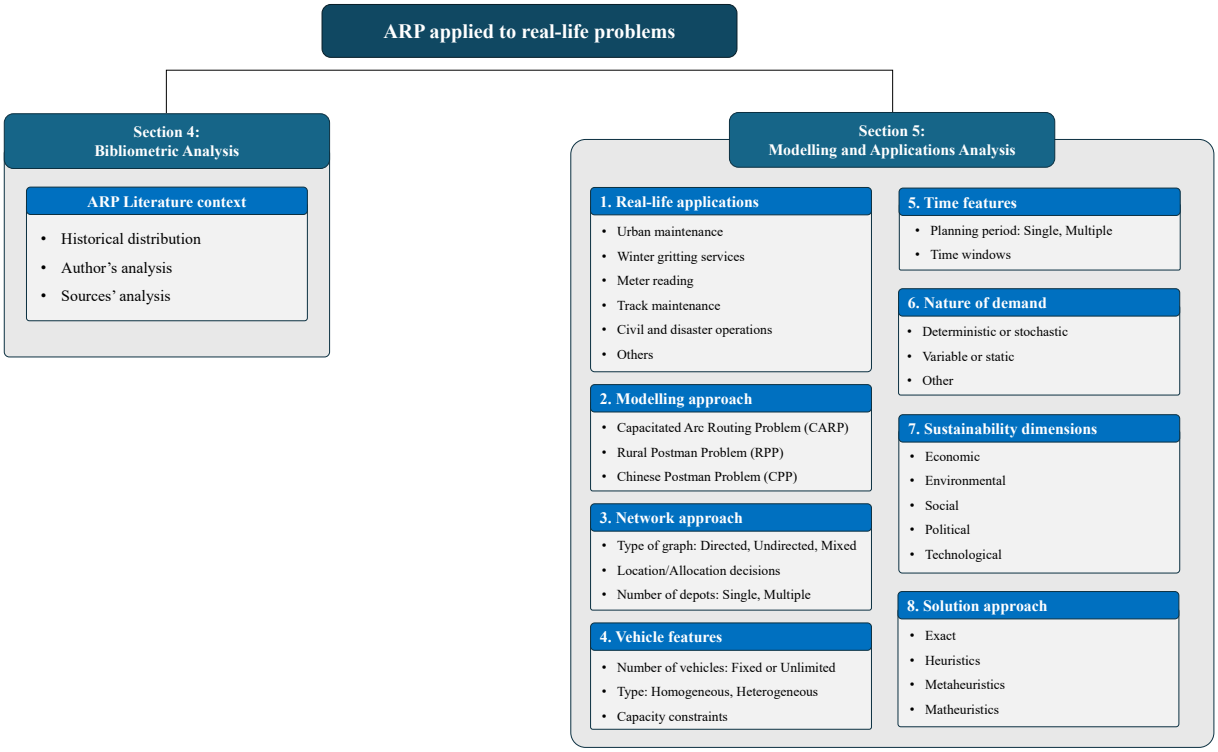


Figure 4: Scheme of the analysis methodology.

4.1. Growth of driving data research

Fig. 5 shows the historical distribution of publications on ARP approaches from 1978 to 2025, with a total of 274 articles. According to the literature found, there is a relatively stagnant trend in research output up to the early 2000s, with fewer than 5 publications annually. A gradual increase in publications began around 2005, with more significant growth observed from 2010 onwards. Notably, there is a sharp increase after 2015, reaching a peak between 2020 and 2023, where the number of publications surges to nearly 25 in a single year. This spike is a five-fold increase in the research activity compared to two decades earlier, highlighting a burgeoning interest and significant advancements in ARP models. This trend analysis indicates a maturing research area with fluctuating but overall increasing attention, emphasising the periods of rapid growth and potential saturation or shifts in research focus in recent years. This summary encapsulates the critical trends and numerical highlights needed for contextualising the ARP literature in a scientific article.

4.2. Most relevant authors and sources

Fig. 6 illustrates the most relevant authors selected in the domain of ARPs and their annual output for the period 2003–2024. The bubble sizes denote the number of annual publications, and the colour intensity is proportional to the total citations per year. For example, MEI Y's name appeared in multiple papers over the period [131], showing consistency in research output. Specifically, MEI Y published three papers in 2010 and 2011 with high citation counts, reflected by darker bubble colours indicating 15 to 20 total citations in those years.
MEI Y, YAO X, and ZHANG M [134] seem to be the top three authors in terms of the total number of articles published, while MEI Y is the most prolific in recent years, since 2019. The group of CORBERÁN, SANCHIS J, and PLANA I [27] shows consistent research activity throughout this period, with a notable impact in 2021, where publications reached maximum citation counts. In terms of the total number of citations, the top five are recorded by CORBERÁN, SANCHIS J, PLANA I (mainly due to their last work), MEI Y and ZHANG M.

The literature dataset is distributed across a total of 145 distinct publication sources. Fig. 7 illustrates the cumulative number of articles from the top five sources. The *Computers and Operations Research* is the most prolific source in the literature, demonstrating a consistent publication trend, with contributions from 2001 to 2016, resuming in 2020 and increasing significantly in 2024, accumulating a total of 23 articles. *European Journal of Operational Research* holds the second position, beginning to publish relevant papers in 2004, with significant growth from 2013 onwards, reaching a total of 20 publications by 2024 in this field. The *Networks* journal is notable for its consistency over time, contributing 16 documents to the dataset. This is followed by the journal of *IEEE Transactions on Evolutionary Computation* and *Lecture Notes in Computer Science* (including subseries in Artificial Intelligence and Bioinformatics), with 8 and 7 publications respectively.

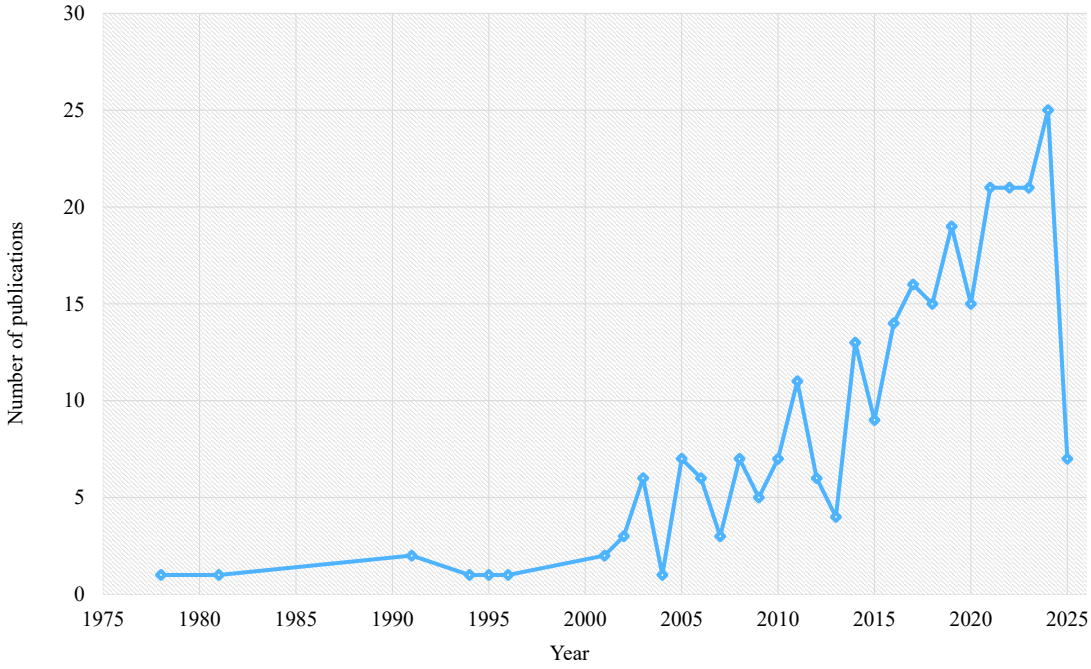


Figure 5: Historical distribution of selected documents.

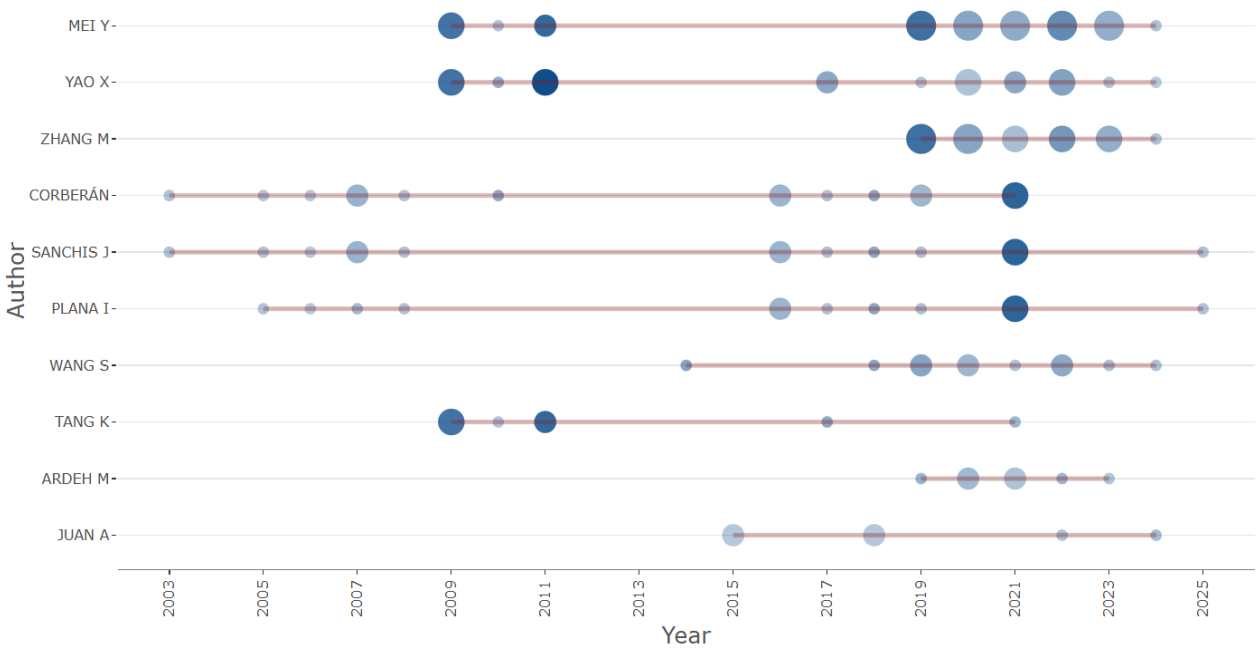


Figure 6: Authors with the highest impact in the scientific domain, based on the number of publications and the total citation count.

5. Modelling and Applications Analysis

The following section presents a detailed analysis of modelling practices and applications in the literature on real-life applications for ARP models. After applying the filters described in the systematic review protocol, the articles were analysed considering different relevant aspects evaluated by the selection criteria discussed in Section 3. The following subsections present the obtained results.

5.1. Real-life applications

Table 2 provides an overview of the analysed documents, categorised according to the real-life application that has been modelled. For this purpose, each application field encompasses a range of case studies grouped by similar activities.

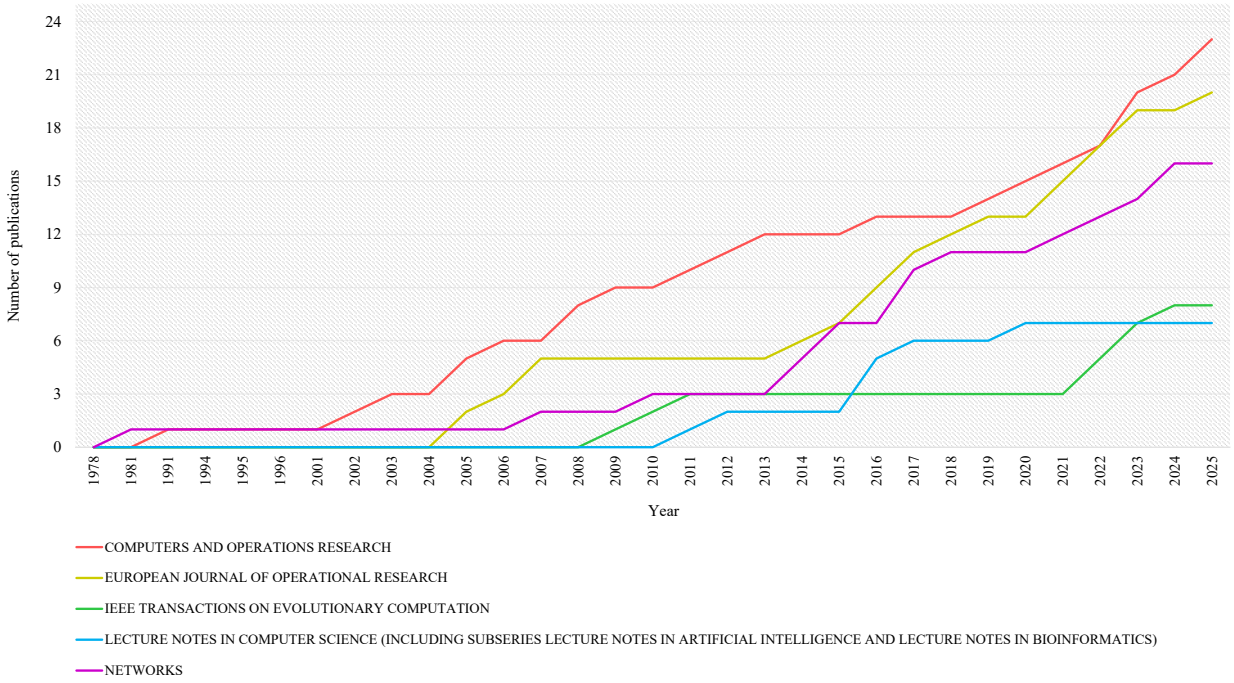


Figure 7: Cumulative growth of scientific literature sources.

These case studies detail the specific activities that each author has addressed in their publications. At the same time, these articles have been organised by the type of graph considered, allowing for an analysis of different perspectives on how to express mathematically each application.

5.1.1. Urban Maintenance

Urban maintenance encompasses several case studies where ARP models are crucial, including road maintenance, street sweeping, truck water sprinkling, and municipal waste collection. While road maintenance applications have been explored using various graph types to reflect different operational strategies, waste collection stands out as the most studied application.

Municipal waste collection has been the most studied application among all fields, being the focus of 26% of the publications. This problem is well suited to ARP models because the task is to collect waste within an urban area, composed of streets with multiple collection points, usually represented by households so that the total demand in each arc of the network is represented by the total waste in each street. This problem has been addressed with directed, undirected, and mixed graphs so far. The prevalence of directed graphs in this field can be attributed to the need for managing one-way streets and designated routes that optimise the collection process and reduce operational times. These studies often highlight how different graph orientations can help manage the complexity of urban logistics by aligning with specific municipal regulations and traffic patterns.

Some problems even consider the routing to disposal sites at the end of the route [119, 127, 100]. Furthermore, some works consider the sorting of different types of waste for proper recycling [132, 50]. In addition to this, the study carried out by [86] assesses the effect of implementing electric vehicles for routing. The work presented in [104] models different situations in waste collection activity, such as a heterogeneous fleet of vehicles, and limitations in the number of visits for waste disposal.

5.1.2. Winter Gritting Services

Winter gritting services are the second most studied application, representing 17% of the results. In this field, case studies focus on salt spreading and snow ploughing, typically employing directed graphs to ensure effective coverage and resource allocation. These activities are critical in maintaining road safety and require precise routing to effectively manage the challenges posed by winter weather conditions. Although the purpose of both tasks is the same, the main difference in terms of modelling is that snow ploughing is a pickup activity in which each side of a road must be treated separately, while salt spreading is considered a delivery activity in which salt can be spread on both sides in a single run. The use of directed graphs helps in establishing routes that maximise coverage while adhering to environmental and public safety standards.

In this field of application is common to identify the use of different types of vehicles. These types of vehicles may differ just in capacity loads [63], or other features like the available fleet, maximum speed and costs [74, 8]. The authors in [45] also studied the use of different trucks for the removal of snow from the driving lanes and the sidewalks.

Field of application	Case study	Directed	Undirected	Mixed
Urban maintenance	Road maintenance	[4], [19]	[118], [52]	[111], [13]
	Street sweeping	[110], [54], [64], [39], [16], [65]		
	Truck water sprinkle	[76], [53], [135]	[76]	
	Waste collection	[127], [123], [10], [129], [56], [100], [92], [109], [119], [86], [87], [49]	[57], [7], [131], [129], [50], [132], [34], [62], [104]	[128], [72] [90], [56], [31], [43], [81], [17]
Winter Gritting Services	Salt spreading	[47]	[91], [85]	[63], [1], [75]
	Snow ploughing	[73], [45], [116], [108], [42], [8]	[55], [51]	[74]
Meter reading	Automated	[14], [102]		
	Conventional	[29], [60],	[122]	[46]
Track maintenance	Metro network	[22], [21], [23]		
	Railway network	[69]	[114], [113]	
Civil and disaster operations	Civil Operations		[105], [3]	
	Debris collection	[98]	[121], [107]	
	Security patrolling		[35], [18]	
	Rescue with Drones	[134]	[134], [2], [115]	
	Traffic monitoring wtih drones	[71], [126], [77], [130]	[126], [89], [25]	[6]
Other	Delivery of goods			[5]
	Agricultural path planning	[61]	[125]	
	Mine dust suppresion		[103]	
	Bus tour planning		[9]	
	Hiking tourism planning	[120]		
	Bank treasury locate	[59]		
	Battery replacement	[117]	[117]	

Table 2: Publications sorted by field of application, case study, and type of graph considered.

In this field of application, it is also common to find the use of Location Arc Routing Problem (LARP) models, sometimes including multiple depots, which include facility location decisions for the designed network. For salt spreading, these decisions are used to locate salt depots in addition to the central depot from which the vehicles depart [55, 91, 85].

5.1.3. Meter Reading

Meter reading has benefited from ARP models mainly through Close-Enough Arc Routing Problem (CEARP), a variation of RPP (see Section 5.2). The use of directed graphs predominates in these studies to minimise travel time and ensure that every meter is visited without redundant paths, enhancing efficiency and reducing operational costs.
As the reading of water and electricity meters is an activity that has become less important due to technological development, some more recent work has studied the case for automated systems for meter reading [14, 102].

5.1.4. Track Maintenance

In the field of track maintenance, particularly within metro and railway networks, directed graphs are utilised to schedule maintenance tasks effectively. These tasks must be strategically planned to minimise disruptions in daily transit operations, and directed nature of the graphs helps in sequencing these activities along the tracks. The work of [114] describes the goal of this task as to find an optimal time schedule for the maintenance activities and optimal routes for the maintenance crew, including all necessary equipment and technicians.
One frequent feature in these studies is the inclusion of time constraints, presented in 65% of the works. Among these works, the main reason for implementing these constraints is to avoid affecting urban passenger transport in the daytime, so the vehicles can only run during nightly time windows. For this purpose, the studies in [21] and [23] also integrate the use of location decisions for intermediate facilities where the vehicles and staff can rest until the next time window is open.

5.1.5. Civil and disaster operations

This field has also been studied in depth, including diverse operations such as debris collection, security patrolling, and drone rescue missions. The directed graphs are especially beneficial in scenarios like disaster response, where routes must be optimised quickly for debris clearance or emergency services to reach affected areas promptly [98, 107, 121]. These applications normally start with an unconnected graph, where vehicles must unblock the damaged roads, allowing the network to be connected once served, providing quick and efficient routing solutions that are vital for effective disaster management.

In recent years, these works also highlight the use of drones in activities such as traffic control [126, 77, 6], security patrolling [89, 25], and humanitarian rescue [134, 2]. The authors in [115] design a model for transportation reconnaissance that proposes temporal and spatial synchronisation in the routing of vehicle and drones. For these purposes, the use of indirect and mixed graphs has been more widely used because they are better adapted to the movements of these aerial vehicles, which normally have no restrictions on the direction of their arcs.

5.1.6. Other applications found

This field includes various other types of applications that could not be classified in other fields.

The authors in [103] study the case of the suppression of mine dust through water sprinkler trucks using an undirected graph. [117] proposed a CARP for the replacement of batteries in an e-bike sharing system, where workers replace batteries for underpowered e-bikes along street segments. The work [9] designs a routing planning model for public transportation, where a single bus picks up AND drops off passengers at multiple stops. [59] proposes a LARP with time windows under demand uncertainty and a heterogeneous fleet to address the problem of the location of treasury centres required for the exchange of money from a bank's branches. The authors in [5] design a competitive transportation-LARP for the delivery of goods with a leader and a follower. The works of [61] and [125] study the case of field path planning for agricultural operations under a CARP approach. The article of [120] develops a sustainable approach under the evaluation of economic, environmental and social objectives for the planning of optimal hiking routes in tourism.

5.2. ARP modelling approaches

The analysed articles can also be sorted according to the model approach studied, starting from the three base variations mentioned in Section 2.2.1: CARP, RPP, and CPP. Then, these variations have been sub-categorised into other variations found in the literature. Lastly, specific models with different names and descriptions have also been proposed. The papers grouped as "Others" correspond to those that cannot be classified in any other category because the authors focus on describing other aspects of their models and not those analysed in this paper. Table 3 summarises the results for this subsection.

The CARP model is the most widely used, present in 72.3% of the total results. This is mainly because having capacity constraints on vehicles allows it to better model a wide range of real-life applications. Although a high percentage of these models do not propose variations (35.6% of CARP models), three sub-categories have been identified in this group:

A) LARP, which integrates facility location decisions such as additional depots in the network configuration,

B) MCARP, which integrates the use of mixed graphs, and

C) PCARP, which uses a multi-period time horizon.

Additionally, a considerable number of articles were considered within the "Others" sub-category, as the description of these models deals with specific aspects of their models or the case study addressed, such as the use of drones [6, 134, 77, 130], particular characteristics for vehicles such as multi-compartment [132, 87] and heterogeneous fleet [74], or the use of intermediate facilities [87, 127, 110, 43].

The RPP has been the second most studied model, representing 11.8% of the studies. The only sub-category found is the CEARP, which accounts for half of the studies on this variant. This model applies to scenarios where the vehicle doesn't need to arrive precisely at the customer's location to deliver its service. Instead, the service can be carried out from a certain distance, requiring the vehicle to merely pass "close enough" to the customer's position [27]. As with CARP models, the "Others" sub-category in RPP presents quite specific model propositions that could not be classified into other groups. For example, they present periodic models [13, 23], inclusion of intermediate facilities [23], and other specifications for each case study.

The CPP has only been identified in four articles. While the works in [39] and [16] respond to street sweeping applications employing a traditional CPP, the work of [111] proposes a hierarchical CPP, described by the authors as the problem to find the shortest tour that shall pass each arc at least once via considering the precedence relations in a hierarchical graph where the arcs are classified according to these precedence relations.

The "Others" category corresponds to articles that do not correspond to any of the three mentioned variations. The Generalised Routing Problem (GRP) developed in [107] combines arc routing and node routing on the same graph. Some other studies are based on the application of Arc Orienteering Problem (AOP) [120, 126]. The majority of other studies name their model according to its application, generalising the model's characteristics in this context (see e.g., [120, 40, 41]).

Finally, it is important to note the large number of results classified as "Others". This is because there is no convention in the literature for naming the designed ARP models, so there may be models that, although having similar characteristics and purposes, are named differently by the authors, making them difficult to classify. A notable example of this is the article that describes the use of drones [6, 134, 77, 130].

ARP Variation	Sub-category	Model proposed	Author's model name	References
CARP				[109], [73], [45], [71], [61], [117], [114], [100], [75] [63], [98], [4] [92], [103], [62] [135], [72], [104], [123], [118], [121] [47], [116], [49], [7], [113]
	LARP		Location ARP	[59], [50], [55], [91], [85]
		CoTLARP	Competitive Transportation-LARP	[5]
		DLARP	Dynamic LARP	[86]
	MCARP		Mixed Capacitated Arc Routing Problem (MCARP)	[56], [90], [81]
		MCARPTC	MCARP with turns constraint	[10]
		MCARPTIF	MCARP under Time Restrictions with Intermediate Facilities	[128], [129]
		MCGRP	Mixed Capacitated General Routing Problem	[1]
	PCARP		Periodic Capacitated Arc Routing Problem (PCARP)	[119], [131]
		PARPRP	PCARP with Refill Points	[53]
		TGCRP-PD	Track Geometry Car Routing Problem with Periodic Demands	[22]
	MDCARP		Multiple-Depot Capacitated Arc Routing Problem (MDCARP)	[17], [65]
		MDHCARP	Multi-Depot Heterogeneous Fleet CARP	[74]
		MDARPFED	Multi-Depot ARP with Flexible Assignment of End Depot	[64]
	Others	AMMDRPG	All Terrain Mothership and Multiple-Drone Routing Problem with Graphs	[6]
		RTOARPM	Robust Drone ARP	[134]
		DL-ARP	Double Layer-ARP	[77]
		GVD-ARP	GV-Drone-ARP	[130], [25]
		DARP	Dissimilar ARP	[26]
		DARP-DD	Drone ARP with Deadheading Demand	[2]
		OCARP	Open CARP	[60], [122]
		RARP	Rich ARP	[76], [21], [69]
		SARP	Sectoring ARP	[31]
		ARPTPC	ARP Time Penalty Costs	[57]
		CARP-VSD	CARP with Vehicle/Site Dependencies	[110], [43]
		CSMC-CARP	Commodity-Split Multi-Compartment CARP	[132]
		MCCARPIF	Multi-Compartment CARP with Intermediate Facilities	[87]
		MLCARPIF	Multi-Level CARP with Intermediate Facilities	[127]
		SyARP	Synchronised ARP	[108]
		CARPMD	CARP with Mobile Depots	[34]
		CARP-SSTT	CARP with Stochastic Service and Travel Times	[20]
		MTCARPTW	Multi-Treatment CARP with Time Windows	[52]
		SD-CARP	Split Delivery CARP	[52]
RPP				[3], [51]
	CEARP			[46]
		DC-CEARP	Distance-Constrained CEARP	[29], [30]
		MM-CEARP	Min-Max CEARP	[14]
		PCEARP	Profitable CEARP	[102]
	Others	PRPP-IS	Periodic RPP with Irregular Services	[13]
		MO-PGDRPPLRIF	Multi-Objective Periodic Generalized Directed RPP under Length Restrictions with Intermediate Facilities	[23]
		TVARP	Target Visitation ARP	[105]
		CBT	City Bus Tour	[9]
		MMMDRPP	Min-Max Multiple-Depot RPP	[18]
CPP				[38], [39], [16]
	HCPP		Hierarchical CPP	[111]
Others		GRP	Generalised Routing Problem	[107]
		HTP	Hiking Tourist Problem	[120]
		TOARP-TP	Team Orienteering ARP with Time-varying Profits	[126]
		TDMVD-TOARP	Time-Dependent Multi-Visit Dynamic Team Orienteering ARP	[42]
		VDCRP-PR	Vehicle and Drone Routing Problem integrated with Point & Arc Routing and Temporal & Spatial Constraints	[115]
		BOARP	Bi-Objective ARP	[35]
		eARP	Electric ARP	[40]
		PERP	Parking Enforcement Routing Problem	[41]
		ARP-SST	ARP with Stochastic Service Times	[19]
		No name given		[15], [8], [89]

Table 3: Publications sorted by model approaches.

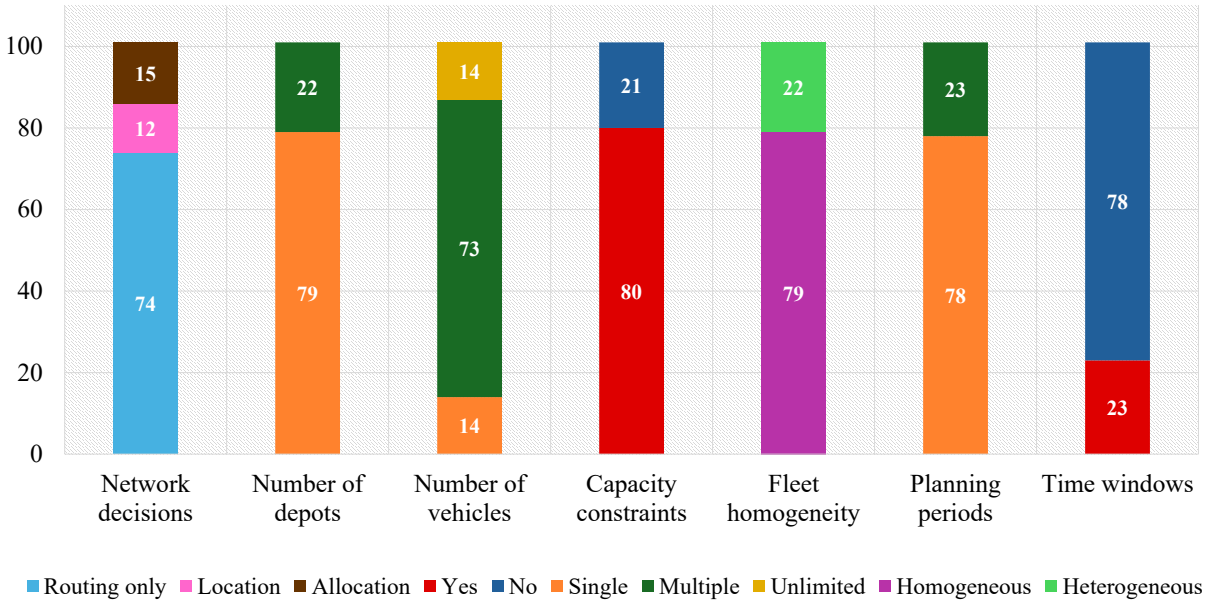


Figure 8: Distribution of publications according to different model features.

5.3. *Model features*

Figure 8 shows the distribution of publications according to different characteristics of the model. In this case, these can be separated into network, vehicle, and time features.

5.3.1. *Network features*

An important feature that characterises the network in modelling is the type of decisions made within it. In this regard, the literature has focused mainly on operational decisions related to vehicle routing, corresponding to 74 articles in this survey. However, it is also possible to identify studies that have included location and allocation decisions.

Location decisions correspond to the strategic level and represent the selection of sites to implement potential facilities, such as depots. These decisions are examined in 12 publications, which take into account the LARP models (7) [55, 50, 85, 86], some multi-depot models (2) [75, 74], and CARPs incorporating the use of intermediate facilities (3) [127, 21, 24]. On the other hand, allocation decisions are tactical decisions which describe the assignment of resources for the facilities, such as vehicles, specialised machinery, or personnel (drivers and assistants), or the assignment of tasks, such as arcs to be served. The results of the survey show that 15 articles have considered allocation decisions, which include only the assignment of vehicles (8) [119, 52, 86] and tasks (7) [51, 3, 85], represented by arcs that need to be served. The works in [5] and [59] include them both simultaneously. The authors in [132] also include the assignment of fractions of waste for an optimal distribution in multi-compartment vehicles.

A second important feature that defines the network is the number of depots considered in the activity. When considering multiple depots, these can already be fixed or can be part of a location decision. This survey has identified 22 multi-depot models. Among these studies, the work in [65] highlights the design of a model in which the starting depot for vehicles is not necessarily the end depot, providing a more flexible assignment. The authors in [17] developed a memetic algorithm to solve a multi-objective, multi-depot problem for the case of waste collection. In [86], a model adapted for electric vehicles is proposed, considering both depots and plug-in and wireless charging stations.

5.3.2. *Vehicle features*

Among the different characteristics presented in the vehicles in the ARP problems, those that have been considered the most relevant for the modelling have been identified. Since most of the articles selected for this review are CARPs or variations of this model, the vast majority of the vehicles have capacity constraints, being present in 80% of the results. This capacity has been modelled for vehicles as the maximum supported load, and for electronic devices such as electric vehicles with charging stations [86, 40] and drones as the maximum battery capacity [115, 2].

Regarding the number of vehicles in the model, two general cases have been identified. Models with a single vehicle and models with multiple vehicles. Works with single vehicles can be found in applications such as meter reading [46, 29, 102], patrolling [105, 126] and rail track maintenance [23]. On the other hand, work using multiple vehicles can be categorised as limited, if there is a fixed maximum fleet present in the model parameters [114, 71, 10, 116], or unlimited if there is no fixed number and it is the model that decides the optimal number of vehicles to use [132, 55, 1, 130].

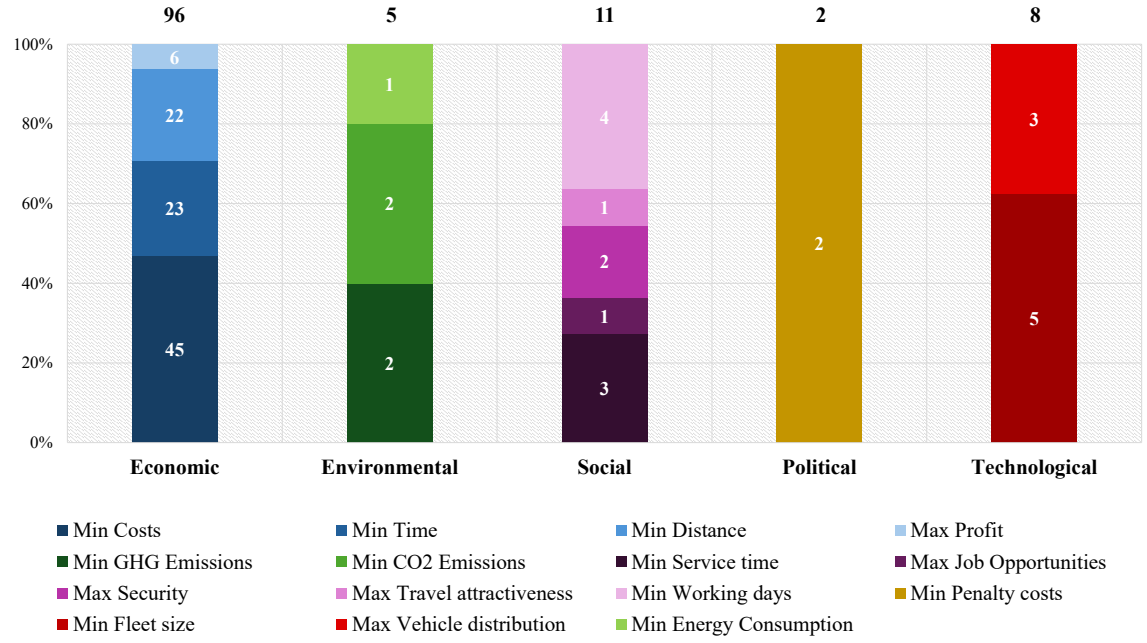


Figure 9: Distribution of publications according to sustainability dimensions considered.

With respect to homogeneity, most of them consider an identical fleet of vehicles. While only 22% of studies evaluate heterogeneous vehicles represented by differences in capacities [127, 63], speed [8], compartment structure [132], and task compatibility [74, 121, 104].

5.3.3. Time features

Different time features have been considered in the literature in order to model the desired field of application better. One of them is the planning distribution, which can be computed over a single period or multiple periods. A single-period planning supposes that all the required arcs are served during the same period. This has been the most common modelling in the literature, used in 78 of the articles reviewed. On the other hand, in multi-period planning, one has to decide which arcs are served initially and which ones are postponed to the next periods [67]. The results show that only 23 publications have used this approach, see e.g., [59, 42, 41, 26, 119, 105].

Time windows have also been identified as key constraints to model real-life scenarios in some operations. While most of them are represented as either working hours [71, 47, 128, 129, 92], or time constraints in which the service must be guaranteed [57, 13, 43, 21], there are also particular time constraints. The work of [59] includes time constraints to reduce the risk of transporting money. Moreover, the authors in [76] represent time constraints as peak hours across a day in a track maintenance application. The work carried out in [115] establishes reconnaissance time windows for spatial and temporal coordination of drones and vehicles.

5.4. Nature of demand

Uncertainty has not been extensively considered in the analysed articles, since more than 86% of the results consider studies under a deterministic view, i.e., the parameters are known with certainty. From the results, only 14 articles consider uncertainty in their parameters, representing 13.9% of the total.

The stochastic models mostly consider the arc demand as an uncertain parameter, measured by a probabilistic distribution [1, 71]. Some of these papers employ robust optimisation for uncertainty modelling [1, 59]. In a similar way, fuzzy models also mostly model uncertainty through demand, but this time using the membership function of a triangular fuzzy number [7, 50].

According to the variability of the demand, some authors also modelled it as a dynamic parameter [117, 126, 42, 98]. Conversely, the work in [42] defines a demand of snow that needs to be removed from arcs and changes each period based on two parameters, the intensity of the snowfall and a fixed threshold of snow that needs to be met.

5.5. Sustainable objectives

Figure 9 represents the distribution of the reviewed publications according to the sustainability dimensions considered in their model objective functions. The economic dimension is by far the most extensively studied, being present in 96 of the 101 articles reviewed (95%). Among these studies, minimising cost is the predominant objective, especially when

it comes only to minimising the transport cost [46, 114, 11, 1], associated with the number of arcs served. However, it is possible to identify studies that also consider other variable costs, such as service costs [57] and various operational costs [71]; as well as fixed costs, including the installation of facilities [59, 55] and the fixed cost of the vehicles used [87, 7, 55]. In addition, minimisation of travel time has been a considerable objective (24%) [6, 77, 31, 89, 49], having some variations like minimisation of makespan [45, 63] and deadhead time [56]. Likewise, distance has been studied significantly, representing 23% of the results. In these studies, the minimisation of travelling distance is the most commonly used [60, 73, 123, 22], followed by the minimisation of deadhead distance [47, 91]. Finally, complementing the evaluation of costs, some articles (6) have also assessed the maximisation of profit, considering revenues related to the served arcs [134, 102, 5].

Regarding the environmental dimension, this has only been evaluated in 5 publications. Some of these works incorporate this dimension in the economic objective function of cost reduction through carbon emission costs [118, 72]. Only 2 recent articles have incorporated particular objective functions for this dimension, such as the minimisation of Greenhouse gas (GHG) emissions [119] and carbon emissions [120, 17]. Furthermore, the authors in [54] propose a multi-objective model which aims to minimise the energy consumed by vehicles.

The social dimension is the second most studied dimension, representing 11% of the total of the studies, and has been addressed through different approaches. From a customer and citizen satisfaction perspective, the minimisation of service times [55, 117], the maximisation of trip attractiveness in tourism applications [120], and the maximisation of safety in cases of patrolling and inspection [130, 42] have been evaluated. On the other hand, from a labour perspective, the maximisation of job opportunities [119], minimising working days [23, 21], and minimisation of workload [3] have been assessed.

The political dimension has been the least considered in these studies. The only way in which this dimension was found was in economic objectives as penalties for non-compliance with service [57, 103].

The technological pillar of sustainability has been assessed in 8 studies and it has focused mainly on the minimisation of the vehicle fleet [52, 131, 53, 21]. In addition, vehicle compactness and distribution have been evaluated in some documents [110, 35].

Finally, it is relevant to mention the lack of multi-objective models under a sustainable approach. From the articles in the survey, 13% adopted a bi-objective model, and only 4% evaluated three objectives [21, 23, 17], with only one work including economic, environmental and social objectives [119]. None of the articles reviewed model notions of regenerative operations or practices.

5.6. Solution methods

Figure 10 illustrates the historical distribution of solution methods employed in the literature on ARP models, spanning several years. Machine Learning has been applied to solution methods such as heuristics, metaheuristics, matheuristics, and decomposition strategies for exact methods. This graphical representation categorises the solution methods into these four distinct types.

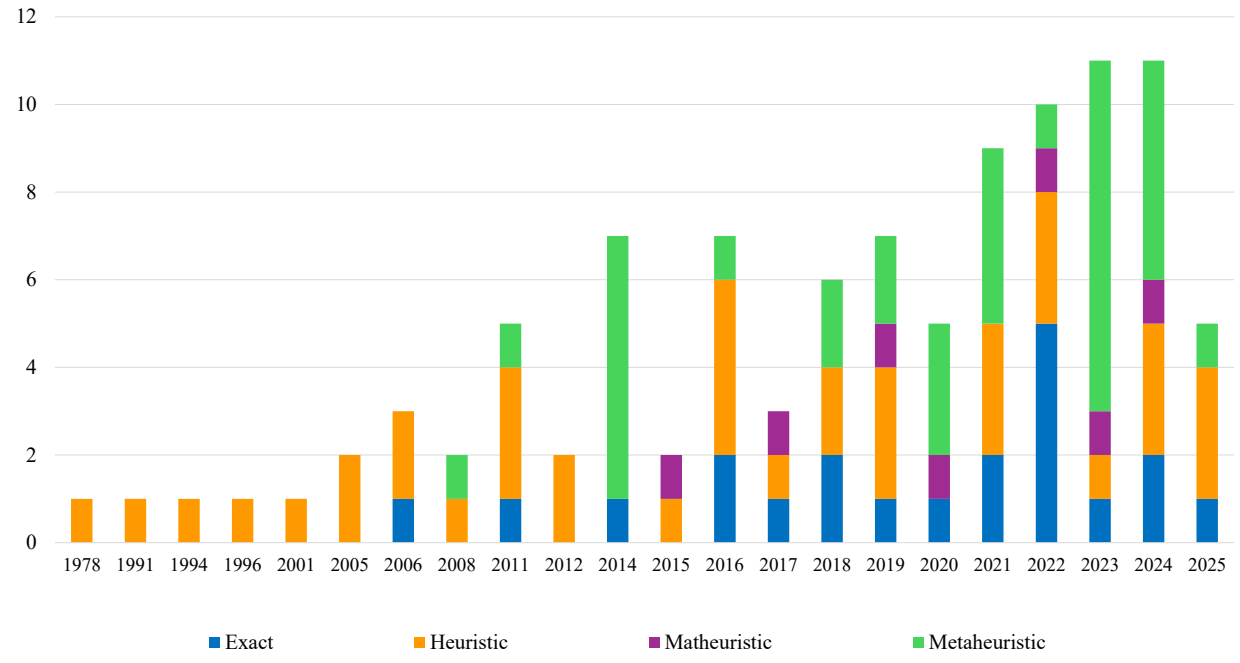


Figure 10: Historical distribution of publications according to the solution method proposed.

A noteworthy trend evident from the chart is the pronounced growth in the adoption of machine learning techniques for developing algorithms such as metaheuristic and matheuristic methods in recent years. These methods have seen a substantial increase, reflecting their rising prominence and effectiveness in addressing complex optimisation in ARP. Among metaheuristics, the most commonly used algorithms are Adaptive Large Neighborhood Search (ALNS) [103, 61, 127, 87], Ant Colony Algorithm [72, 53, 109, 60], Genetic Algorithms [86, 118, 135], and Memetic Algorithms [22, 131], each of these with their respective variations. Regarding matheuristics, [29] proposed a Constructive Algorithm based on the metaheuristic Local Search Algorithm and the exact method Branch and Cut. The work of [56] provides matheuristics using a Mixed Integer Linear Programming (MILP) formulation and a Geography Information System (GIS) to decrease the size of the MILP problems to solve. The authors in [132] developed a Three-phase Matheuristic Algorithm using a compartment assignment algorithm for choosing the type of vehicle, a vehicle selection phase using ALNS, and an algorithm that selects the set of best routes for the fleet obtained.
Conversely, the use of exact methods has remained relatively stable, indicating a consistent but limited applicability due to the complexity and size of problems. In this field, although some studies solve small instances directly using solver software [59, 114, 9], the most popular algorithm methods are the Branch and Bound [105, 120], Branch and Cut [46, 14], Branch and Price [134, 76], and Column Generation [4, 69].
Heuristic methods, while still employed, show a slight decline in usage over time, possibly overshadowed by the more sophisticated metaheuristic and matheuristic techniques. Among the most commonly used are Local Search heuristics [129, 31, 126, 92], constructive heuristics [128, 10, 89], and various two-stage heuristics [77, 13, 43].
This shift highlights a significant evolution in the field, suggesting a move towards more flexible and powerful solution strategies that can better handle the dynamic and intricate nature of modern arc routing problems.

6. Key Insights into ARP for Roadside Management

According to the analysis based on various evaluation criteria, it is possible to identify models with characteristics similar to those required for roadside management (see Section 2.3). This section aims to compile all the relevant information from the analysed articles. Table 4 presents a comparison between the roadside management requirements highlighted in this work and articles from the literature, sorted by applications, with similar model features. The column "% Literature" shows the total percentage of articles that considered each feature in their models.

Criteria	**Roadside Management requirements**	Waste collection			Snow ploughing				Debris collection	Road maintenance		Track maintenance		**% Literature**
		[17]	[119]	[50]	[74]	[45]	[85]	[8]	[121]	[4]	[52]	[23]	[21]	
Type of graph	Directed with multiple types of arcs					x			x		x			4,95%
Type of decision (excluding routing)	Strategic: Location			x	x		x					x	x	8,91%
	Tactical: Allocation of tasks	x	x	x			x		x	x				7,92%
	Tactical: Allocation of vehicles		x					x			x			7,92%
	Operational: Worker's scheduling													**0,00%**
Number of depots	Multiple fixed depots	x		x	x		x	x	x			x	x	14,85%
Type of vehicles	Heterogeneous fleet: Costs				x		x			x	x			4,95%
	Heterogeneous fleet: Time					x		x		x	x			3,96%
	Heterogeneous fleet: Tasks					x			x		x			4,95%
Time features	Multi-period planning		x									x	x	11,88%
Sustainability	Holisitic approach	x	x										x	2,97%
	Regenerative operations													**0,00%**
Demand	**Varies according to the objective**													**0,00%**

Table 4: Summary of key findings in ARP literature for applications in roadside management.

In this analysis, different applications with similar characteristics have been identified that could be applicable in the context of roadside management activities. The most critical task in road maintenance is mowing, which, like the pruning activity, can be modelled analogously to models where the task is to collect distributed demand on the arcs. Examples of these applications are waste collection and snow ploughing with collection models, as they are primarily based on planning the routing of trucks equipped to collect high volumes of demand.
Complementing these activities, track and road maintenance models also present similarities in the technical part of the maintenance, especially those focused on a railway network. In this type of maintenance, a technical team with staff prepared for different maintenance activities is deployed across the network.
Type of graph: Regarding the design of the network for road maintenance problems, the main gap found in the analysed literature is that they are primarily focused on applications in urban areas such as cities and residential areas (as is the case with municipal solid waste and meter reading), whereas roadside management must be carried out in rural areas. The only similar approach in network design is the applications in railway networks. The infrastructure of roads is different from that of streets (see Section 2.1.1). Unlike streets, roads in rural areas typically have a greater extension, which affects the distance of the arcs that must be traversed. In addition, urban sectors such as cities and residential areas have a higher number of intersections, which translates in the graph as a higher number of vertices, and therefore a higher number of

edges and arcs.
Based on graph theory and the rural road infrastructure, the most suitable type of map for modelling roadside maintenance problems is the directed graph, as maintenance must be done on both sides of the road. However, one of the most distinctive features in roadside management is the segmentation of the arcs according to the different sections of the roadside, which have different maintenance requirements. In this matter, the literature has studied multiple types of arcs that need service, such as lanes and pavement in snow ploughing [45] and roads blocked with different types of debris that require specific vehicles to remove them [121]. Nevertheless, they are modelled separately as different arcs. In contrast, the different sections of roadsides are contained within the same arc. This type of arc has not been found in the literature.
Type of decisions: LARP models can be useful to define the location of depots and collection or valorisation centres for biomass recovery in the roadside management network. Once the biomass collection trucks have reached their maximum capacity or have completed the maintenance route and returned to the technical centre, they must head to the collection centres to valorise the collected biomass. For this activity, applications in waste collection can be useful, especially those that consider routes to waste recycling or final disposal.
Facility allocation is also a key decision in real-life applications. In the case of roadside maintenance, the budget is limited, so the number of tasks assigned to each depot must be determined by the proper allocation of vehicles and workers. Although there is plenty of literature that addresses tasks and vehicle allocation, there is no article studying the scheduling of the available workers at an operational level. Although complicated, this decision is essential for modelling not only roadside management activities, but also a wide range of real-world applications. The literature should explore this field in greater depth.
Number of depots: As with location decisions, LARP models are useful for routing vehicles through multiple depots. There is considerable literature about LARPs for winter gritting services, especially through the use of multiple intermediate facilities.
Type of vehicles: The characteristics of the vehicles are an important point to consider in modelling roadside management, as the different activities performed require different machinery, so it is necessary to consider a heterogeneous fleet of vehicles. In this case, the literature has considered multiple useful approaches, particularly in snow ploughing and road maintenance applications, such as differences in vehicles' costs, service times, and tasks to carry out.
Time features: These can be employed in different ways depending on the interests of the model study. The main functionality in roadside management is the consideration of multiple planning periods since each activity must be done during a certain period of the year due to seasonal factors. Literature has explored this topic in depth, especially in waste collection and track maintenance.
Nature of demand: The demand has been frequently modelled as an uncertain parameter under stochastic and dynamic data. In the context of road maintenance, although the vegetation in green spaces on the roadside depends on uncertain parameters such as the season and the type of vegetation, the feature that differentiates it from other literature applications is that the demand of the model is defined as a certain volume of biomass that depends on the decision of how much to cut according to the objective that is being optimised. For example, if the objective is the minimisation of risks of accidents, decisions should prioritise a low level of vegetation in winding roads, where visibility is poor. This survey has shown that there is no article addressing the demand through this approach.
Sustainability modelling: The development of sustainable roadside management is a growing challenge due to climate change and the growth of rural infrastructure in the coming years. Therefore, a holistic perspective focused on the five pillars of sustainability would allow for a better evaluation of sustainable development and the performance of the multiple ecosystem services that roadsides can provide. Despite this, a second and considerable gap in the literature lies in the scarcity of sustainable objective assessments, focusing mainly on the economic area. However, it is possible to find works like that of [119], who has included economic, environmental, and social objectives in his analysis. Sustainable objectives present in the literature, such as minimising greenhouse gas emissions (environmental) [120], minimising working days (social) [23], and minimising the fleet of vehicles (technological) [53], can be useful in the context of roadside maintenance. Despite these findings, in order to better evaluate a model adapted to roadside management, it is necessary to include regenerative operations that can enhance the performance of ecosystem services provided by roadsides or to apply life cycle assessment through other impact categories such as energy consumption, resource depletion, and human toxicity potential.
Solution methods: All these technical and modelling considerations translate into a model with a complex structure in terms of optimisation, that is, an NP-hard model. For this, multiple solution methods including machine learning techniques have been identified in the literature. Roadside management includes operational decisions that must be made in the short term, so low-resolution times are required. Because of this, the resolution method considered will depend on the complexity of the model and the size of the instance used, needing to find a balance between the resolution time and the distance between the optimal solution and the feasible solution found. One option to reduce the complexity is applying two-stage modelling, separating the maintenance tasks into different optimisation models, where the outputs of the first model become the input for the second optimisation. This reduces the number of variables and the size of the graph since not all arcs require the same type of maintenance, that is, a route may need mowing but not necessarily need pruning. Depending on the case study, there will be limitations with this methodology, as to ensure the models are consistent with each other, it must be ensured that each considered activity serves different arcs, that is, that the same arc is not served twice.

Finally, sustainable roadside management must be evaluated under different sustainable objectives, which implies the use of multi-objective solving methods. The literature has studied almost only bi-objective approaches, leaving a considerable gap regarding multi-objective solution techniques that can cope with more than 3 objectives.

7. Conclusions and Perspectives

The ARP has emerged as an essential framework for optimising logistics in various real-life applications. This paper presents a systematic review of the published literature discussing ARP models applied to real-life problems, aiming to identify and evaluate existing models and their applicability in sustainable roadside management. The existing contribution towards the subject is analysed based on relevant criteria, such as field of application, network approaches, model features, nature of demand, sustainability objectives, and solution methods.

Findings
Several key findings have been identified in the development of this review. The bibliometric analysis revealed a significant increase in the number of publications, indicating growing interest and development in this scientific domain. One of the main drivers for this growth was the development of machine learning techniques by the use of metaheuristic and matheuristic methods, which have the ability to handle complex problems and large instances, overcoming the limitations of traditional exact and heuristic methods.
The analysis of the selected criteria has provided a comprehensive overview of the current state of ARP research. This has allowed for a detailed description of the specific needs for a model adapted to roadside management. Regarding real-life applications, the ARP models have been applied in a wide range of contexts demonstrating their flexibility and adaptability to different logistical scenarios. Particularly, applications in waste collection, winter road maintenance, and track maintenance were found to be useful as examples for modelling roadside management scenarios. Considering the aspect of sustainability, although it was found that the literature mostly dealt with the economic dimension under a mono-objective approach, works that integrated the other dimensions of sustainability as a whole, in particular the technological dimension, were also considered. Interesting applications have also been found in LARP models for biomass recovery, the use of heterogeneous fleets to adapt to the different technologies present in management, differences between urban and rural networks, and applications of uncertainty mainly in demand.

Gaps
Despite the advances that have been made, several gaps in the existing literature limit the potential of ARP models. There are no studies that directly address roadside management using this approach, which highlights the need to develop specific models for this purpose. Although some important features for roadside management have been addressed in the literature through other applications, key components regarding the type of arcs, demand, operational decisions and sustainable and regenerative aspects are still missing. Despite being a major factor in enabling more reliable simulation of many day-to-day operations, operational decisions beyond vehicle routing, such as scheduling workers have not been studied. Additionally, the integration of sustainability via environmental, social, political and technological objectives into ARP models remains limited, indicating the need for more holistic approaches. Furthermore, current models lack comprehensive strategies for incorporating regenerative practices.

Future work
To bridge these gaps, future work should focus on investigating and developing ARP models specifically designed for roadside management, considering both the technical and logistical aspects of maintenance operations. It is also crucial to expand the inclusion of sustainable objectives and regenerative operations in this subject. In the case of roadside maintenance, it is suggested that metrics and methodologies be developed to evaluate the ecosystem services provided by roadsides and how these can be optimised through sustainable management. A promising direction is to include decisions that enable operations to be integrated with nature, minimising harm and even generating positive outcomes on environmental and social systems. Regarding solution techniques, exploring and applying advanced multi-objective optimisation methodologies that can handle the complexity and size of ARP problems in roadside management, such as robust optimisation approaches and hybrid algorithms, is also fundamental.
This article has conducted an exhaustive review of 101 publications on ARP focused on real-life applications. Since this domain is expected to increase, it is recommended to update the search when more results are available. The use of another article selection methodology is advisable in order to find other useful articles in the literature that have not been considered in this work. In addition, it is also suggested to perform new analyses such as a more extensive bibliometric analysis, as well as to further study the relationship between the proposed evaluation criteria, providing a wider range of ways to organise and display driven data. Finally, the ARP has the potential to be applied to a wide range of real-life problems, and it is therefore recommended that similar methodologies be applied to other fields of application that have not been studied in the literature.

Declaration

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Acronyms

ALNS Adaptive Large Neighborhood Search. 16

AOP Arc Orienteering Problem. 13

ARP Arc Routing Problem. 2–14, 16, 17, 19

CARP Capacitated Arc Routing Problem. 5, 8, 12, 14

CEARP Close-Enough Arc Routing Problem. 11, 12, 14

CPP Chinese Postman Problem. 5, 12–14

GHG Greenhouse gas. 15

GIS Geography Information System. 16

LARP Location Arc Routing Problem. 11, 12, 14, 18, 19

MCARP Mixed Capacitated Arc Routing Problem. 12, 14

MILP Mixed Integer Linear Programming. 16

PCARP Periodic Capacitated Arc Routing Problem. 12, 14

PRISMA Preferred Reporting Items for Systematic Reviews and Meta-Analyses. 3, 5

RPP Rural Postman Problem. 5, 11–14

VRP Vehicle Routing Problem. 2, 4, 9

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